EXPERIMENT NO. 8

CONFIGURATION OF PATH VECTOR ROUTING PROTOCOL (BGP)

AIM

To study and implement BGP for inter-domain routing.

OBJECTIVES

- To configure BGP with different autonomous systems.
- To establish neighbor relationships.
- To study BGP path selection attributes.
- To analyze BGP routing table entries.
- To verify communication between different AS networks.

THEORY

Border Gateway Protocol is a path vector routing protocol used for routing between autonomous systems. It is widely used for inter-domain routing.

BGP uses attributes such as AS Path, Next Hop, Local Preference, Weight and MED to select paths and prevent loops.

SOFTWARE / MATERIALS REQUIRED

- Cisco Packet Tracer 8.x
- Computer System

IMPORTANT COMMANDS / PROGRAM STEPS

```
router bgp 100
neighbor 10.10.10.2 remote-as 200
network 192.168.1.0 mask 255.255.255.0
show ip bgp summary
show  ip  bgp
ping 192.168.2.2
```

PROCEDURE

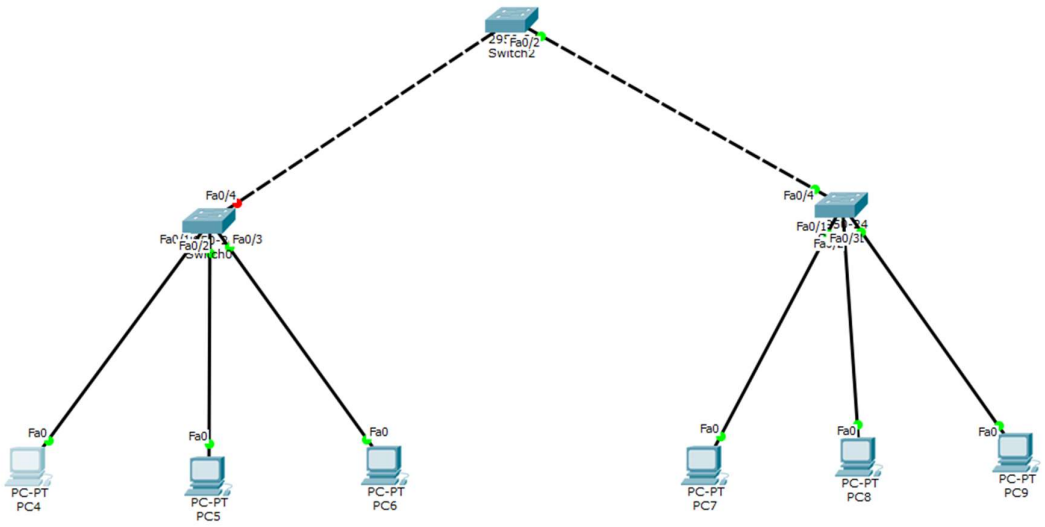
1. Configure router interfaces and IP addresses.
2. Assign different autonomous system numbers.
3. Configure BGP neighbor relationships.
4. Advertise local networks using the network command.
5. Verify BGP neighbors and routing table.
6. Test communication between different AS networks.

OBSERVATION / COMPARISON TABLES

Parameter	Observation
Autonomous System Number	100,200,300
Neighbor Status	Established
AS Path	100 200 300
Routes Advertised	10.10.10.1
Routes Received	10.10.10.1, 20.20.20.1, 30.30.30.1
Ping Response	Success

Attribute	Purpose
AS Path	Loop Prevention
Next Hop	Destination Router
Local Preference	Preferred Exit Path
MED	Route Selection Metric

RESULT



Conclusion

BGP is a routing protocol used to exchange routing information between different Autonomous Systems. The Neighbor Status shows whether the BGP connection is successfully established, while the AS Path shows the sequence of Autonomous Systems through which a route has passed. Routes Advertised are the networks sent by a router to its BGP neighbor, and Routes Received are the networks learned from the neighbor. Together, these parameters help network administrators monitor BGP connections, understand routing paths, and ensure efficient and reliable communication between networks**.**